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13 May 1953
SUPERSEDING
Army 51-75-70
30 November 1948

MILITARY SPECIFICATIONS

MOUNT, TELESCOPE, M30

This specification consists of this cover sheet and Specification 51-75-70, dated 30 November 1948 attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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U. S. ARMY
SPECIFICATION

No. 51-75-70
30 NOVEMBER 1948

MOUNT, TELESCOPE, M30

A. APPLICABLE SPECIFICATIONS

A–1. Ordnance Department Drawing Class 90, Division 91, Drawing 1, titled "Mount, Telescope, M30, List of Drawings," lists or indicates all drawings and specifications which form part of this specification.

A–2. All drawings and specifications shall be of the issue in effect on date of invitation for bids. The issue in effect shall be that issue which the contracting officer shall designate or furnish the bidder.

B. TYPES

B–1. This specification covers one type of instrument known as "Mount, Telescope, M30."

C. MATERIAL AND WORKMANSHIP

C–1. *Material* shall be in accordance with the drawings, general specifications, and material specifications forming part of this specification.

C–2. *Workmanship* shall be in accordance with U. S. Army Specification No. 51–7, Instruments, Fire Control, General Specification Governing the Manufacture and Inspection of.

D. GENERAL REQUIREMENTS

D–1. *General requirements* and associated tests common to all fire control instruments are contained in U. S. Army Specification No. 51–7 and are mandatory in so far as they are applicable.

E. DETAIL REQUIREMENTS

E–1. *Adjustment.*—With the mount set up as shown on the assembled views drawing, with the mounting surface of the housing in a vertical plane, with the leveling pads of the actuating arm horizontal, with the axis of the telescope socket vertical, and with a properly adjusted panoramic telescope in place in the socket; the mount shall be adjusted as follows:

E–1*a*. *The elevation level vial* shall be adjusted to show the bubble central with respect to the level vial graduations.

E–1*b*. *Collimation.*—The adjusting screws in the telescope socket shall be adjusted to bring the line of sight of the telescope parallel to the mounting surface of the housing. In this position, the locating lug shall be approximately in the center of the adjustment.

E–1*c*. *Cross level.*—The mount shall be adjusted by turning the cross leveling knob to the position where the deviation of the line of sight of the telescope from a vertical line on a target is minimum when the actuating arm and the hub attached to it on the

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trunnion are rotated from 0° to 65° elevation about an axis perpendicular to the mounting surface of the housing. In this position the cross level vial shall be adjusted to show the bubble central with respect to the level vial graduations.

E–2. *Functioning.*—All moving parts shall function properly without irregularities, undue friction, or undue looseness.

E–3. *Accuracy.*

E–3*a*. When the mount has been set up and adjusted as in paragraph E–1, it shall meet the following requirements:

E–3*a*(1). *The axis of the telescope socket* shall be vertical in the cross level plane within 1 mil (3 min., 22.5 secs.).

E–3*a*(2). *The top surface of the telescope socket* shall be horizontal within 0.3 mil (1 min.). (This surface is made horizontal to provide means of adjusting the level vials in the field.)

E–3*a*(3). The deviation remaining after adjustment as in paragraph E–1*c* shall not exceed 0.5 mil (1 min., 41 sec.) either vertically or horizontally.

E–3*a*(4). The line of sight of the telescope shall follow a plumb line within 0.5-mil (1 min., 41 sec.) when the vertical travel of the telescope and mount is accomplished by means of the elevation worm through the limits of the elevation worm movement.

E–3*a*(5). The line of sight of the telescope shall remain parallel to the mounting surface of the housing within 1 mil (3 min., 22.5 sec.) when the extension is placed in the socket and the telescope properly mounted.

E–3*b*. *Azimuth correction.*—With the mount attached to a surface which is at an angle with the vertical, and with the telescope socket leveled, turning of the actuating arm and the hub attached to it on the trunnion end with respect to the housing around an axis perpendicular to the mounting surface shall cause the line of sight to turn in azimuth according to the formula and table below:

$$\tan \lambda = \tan \psi^1 \sin \epsilon$$

ϵ = crosslevel angle (angle between plane of mounting surface of housing and vertical)

ψ^1 = angle through which hub and arm have been turned (similar to elevation angle of gun)

λ = azimuth correction (horizontal angle through which line of sight has moved)

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Cross level angle (ϵ) right or left (degrees)	Elevation angle (ψ) (mils)	Azimuth correction (λ) (mils)	Tolerance on azimuth correction (mils)
5 (88.89 mils)	300	26.9	± . 7
5 (88.89 mils)	600	59.2	± 1.1
9 (160 mils)	300	48.3	± .9
9 (160 mils)	600	106.0	± 1.6

E–4. *Shock.*—After being subjected to ten 300-foot pound shocks, the mount shall not be damaged and it shall meet the requirements of paragraph E–3.

E–5. *Temperature* requirements and test shall be in accordance with Type III as specified in U. S. Army Specification No. 51–7.

F. METHODS OF INSPECTION AND TESTS

F–1. Unless otherwise specified herein, each mount and extra part shall be inspected for conformance with this specification and other applicable specifications and shall be subject to inspection during process of manufacture and assembly as specified in U. S. Army Specification No. 51–7.

F–2. Each mount shall be tested for conformance with the requirements of section E.

F–3. The azimuth correction test required of this mount is performed as follows:

(1) Attach the mount, with a panoramic telescope in place, to a suitable fixture so that the actuating arm and its attached hub may be rotated about an axis perpendicular to the mounting surface of the housing while this axis is maintained at an angle with the horizontal equal to the required cross level angle (ϵ).

(2) Level the body assembly as indicated by the level bubbles.

(3) At zero elevation, direct the line of sight of the panoramic telescope at a plumb line target.

(4) Rotate hub assembly (C79356) and arm assembly (C79357) through an angle equal to the required angle ψ^1 measured in the plane of the mounting surface of the housing.

(5) Measure on the azimuth circle of the panoramic telescope the azimuth correction (λ) through which the head of the panoramic telescope must be turned to bring the line of sight back to the plumb line.

(6) Repeat, with the cross level angle applied in the opposite direction.

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F–4. *Shock test.*—This test shall be made on an impact testing fixture in general accordance with Ordnance Drawing D7693860. Using a suitable adapter, the mount shall be mounted to simulate the "on-carriage" position (see note 1). The direction of the impact shall simulate the shock received by the mount because of the normal recoil of the gun.

G. PACKING AND MARKING

G–1. Packing of the telescope mount shall be in accordance with U. S. Army Specification No. 51–7 and supplementary instructions supplied by the contracting officer.

G–2. Packaging of spare parts shall be in accordance with Ordnance Department Packaging Instructions No. AFPS–3 and/or AFPS–41–1.

G–3. Marking for shipment shall be in accordance with U. S. Army Specification No. 100–2.

H. NOTES

H–1. It is recommended that a dummy panoramic telescope or an object of equivalent weight and c.g. with extension C80205 (both furnished by the Government) mounted be used when conducting this test.

H–2. This specification replaces Frankford Arsenal Tentative Specification FXS–445.

H–3. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Department of the Army.

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any purpose except when specifically authorized by the Chief of Ordnance.

Note.—Copies of this specification may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

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